Purpose of case hardening

1. To provide adequate wear resistant on the surface.
2. To improve corrosion resistance
3. To improve heat resistance
4. To increase life of components made from low cost material.

The different processes of case hardening are

1. Carburizing
2. Nitriding
3. Cyaniding
4. Induction Hardening
5. Flame Hardening

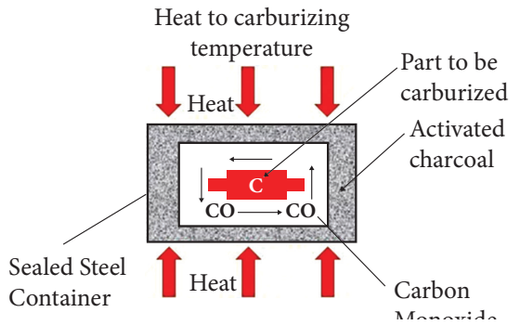
CARBURIZING

There are several techniques for hardening the outer surfaces of steels. These include diffusing carbon into the surface and heat treating the surface without changing the properties of core. This treatment is applied to low carbon steel parts after machining

There are three types of carburizing commonly used:

1. Pack Carburizing
2. Liquid Carburizing
3. Gas Carburizing

Pack Carburizing: In pack carburizing, the components are surrounded by a carburizing medium and placed in a sealed box. The medium is usually wood (or) leather, coke or charcoal mixed with barium carbonate. When the sealed box is kept into the furnace at 925°C and held at the same temperature for considerable time. At that temperature carbon monoxide is produce from the medium and reacts with the iron. It produces iron carbide on the surface. It hardened the surface.



Pack Carburizing

where C Means Carbon on the Surface of the work piece and CO Means Corbon Monoxide gas that is circulated around the work piece.

Disadvantages of Carburizing

1) It is not suitable for mass production.
2) Time taken is more for packing, unpacking, heating and cooling.
3) The amount of hardening cannot be controlled easily.

NITRIDING

Nitriding is one of the case hardening process on alloy steels

In this process nitrogen is added to the surface of steel parts. The steel parts is heated in the presence of Ammonia gas at range of temperature 500°C to 600°C The Ammonia is dissociated as Nitrogen and Hydrogen, the nitrogen reacts with the steel to form nitrides. It develop a very hard case in a component at low temperature. In this process the surface is hardened up to a thickness of 0.25mm.

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